

Chapter 9:

Mechanisms of evolution and speciation

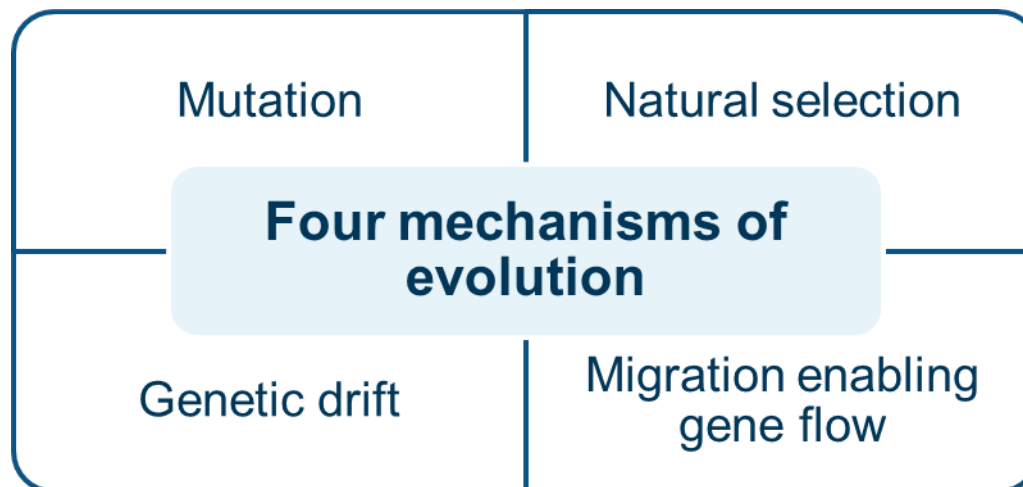
Evolution and its mechanisms

Terminology

Evolution is the process of cumulative, heritable change in a population over many generations.

The **theory of evolution** states that all organisms have developed from previous organisms and that all living things have a common ancestor in some initial form of primitive life.

Mechanisms cause the basis of evolution; changes in allele frequencies within a gene pool.



A mechanism for evolution: mutation

Mutation

Mutation is a source of new alleles in a population's gene pool.

A mutation is a permanent change in the DNA sequence of a gene. A mutation can change one allele into another, but the net effect is a change in the frequency of an existing allele.

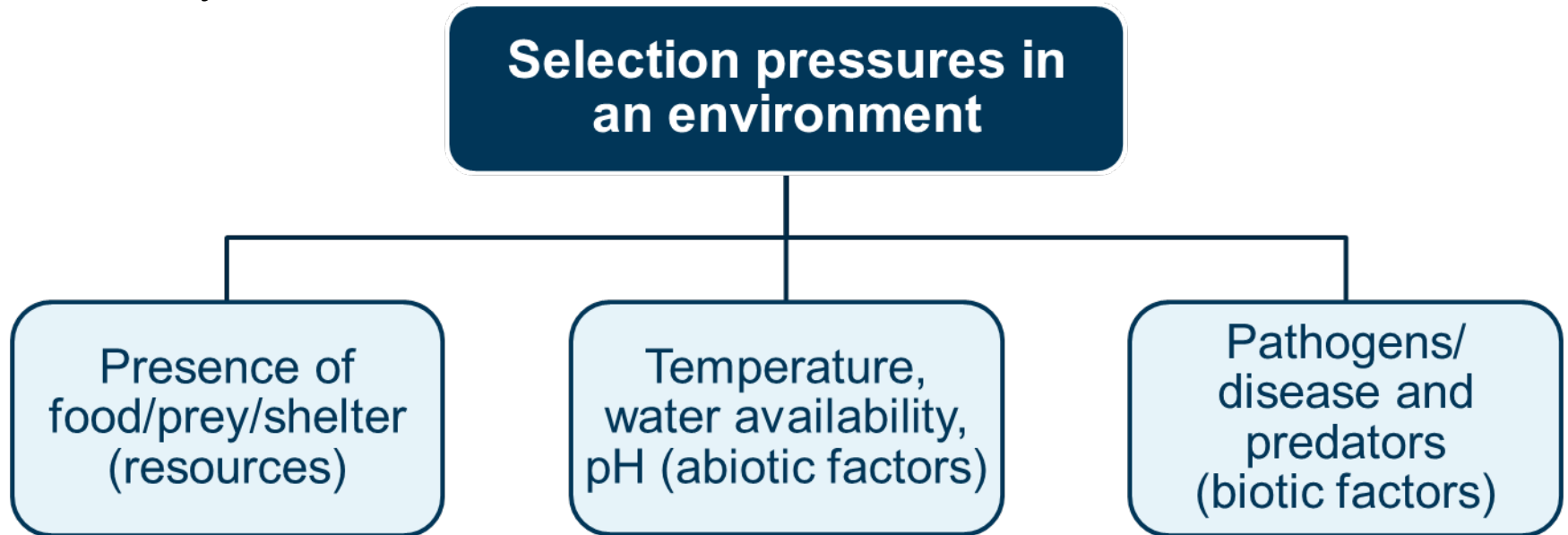
The change in frequency resulting from mutation is small, so its effect on evolution is insignificant unless it provides a beneficial trait with respect to a particular selection pressure in the environment.

A mechanism for evolution: mutation

Selection pressure

A **selection pressure** is an environmental factor that can be survived by those individuals in a population who possess a beneficial trait, but not others. It can contribute to changes in allele frequency in a population gene pool.

A mutation may produce an allele that is selected against, selected for, or selectively neutral.



A mechanism for evolution: mutation

Selection pressure example:

The peppered moth, *Biston betularia*, has a white speckled *typica* form and a dark *carbonaria* form. Blackening of tree trunks by soot presented a new environmental pressure for the moth population. The dark-coloured moths were better able to evade bird predation than the common white speckled form. Over time, black moths came to dominate the population.



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A mechanism for evolution: mutation

Mutation is the ultimate source of variation

A **population** is a group of individuals of the same species that live in the same geographic area and readily interbreed to produce fertile offspring.

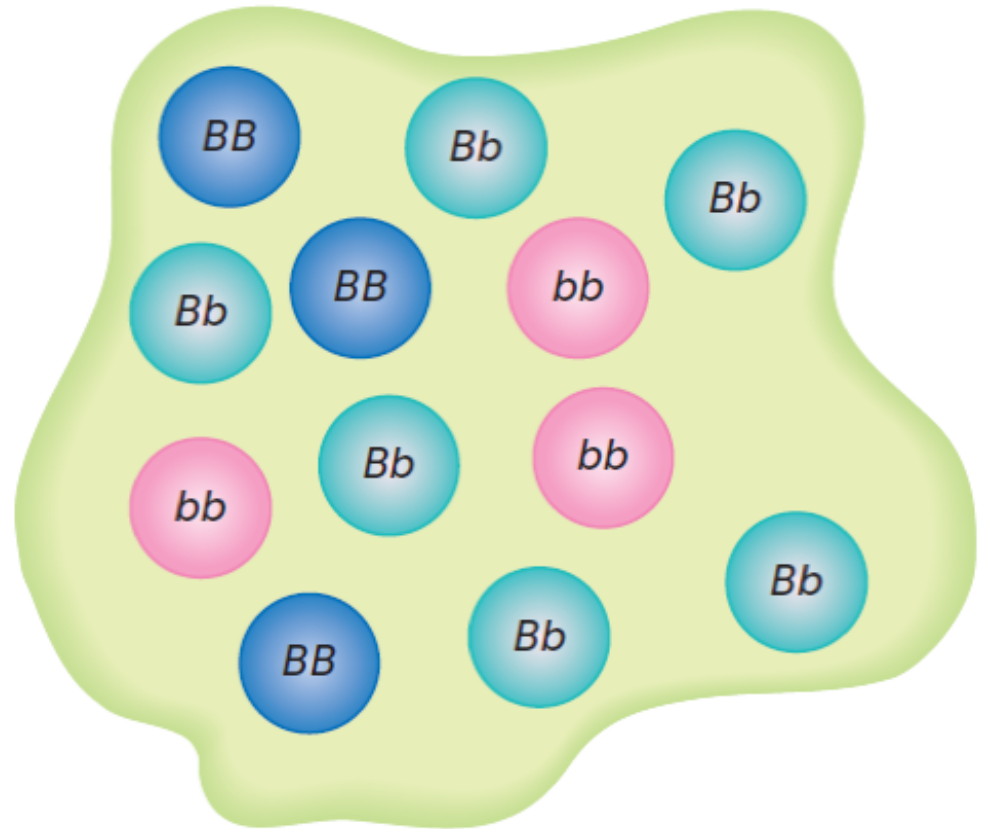
Members of a population have variation in their genotypes that causes variation in their phenotypes.

Variation, therefore, is based on differences in DNA sequences, which give rise to different forms of genes (alleles), which in turn result in different phenotypes.

A mechanism for evolution: mutation

Gene pools

Genes are the means of transmitting phenotypes from one generation to another. Many genes can exist in different forms as alleles, and the characteristics of individuals are determined by the alleles they inherit. It is this variation in alleles carried by different individuals that leads to most of the variation in a population. The total collection of alleles within a population is referred to as the **gene pool**.



A mechanism for evolution: natural selection

Natural selection

Natural selection occurs when selection pressures in the environment confer a selective advantage over a specific phenotype to enhance its survival and reproduction.



It leads to changes in allele frequency in the gene pool of a population.



Individuals that have certain inherited traits are more likely to survive and reproduce at higher rates than other individuals.

A mechanism for evolution: natural selection

Natural selection principle 1: variation

Individuals in a population differ from one another; that is, individuals within populations show variation.

Variation is due to mutation in alleles and meiosis/sexual reproduction processes. These processes include crossing over, independent assortment and random fertilisation, and random mating.



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A mechanism for evolution: natural selection

Natural selection principle 2: overproduction

There are more individuals produced in a population than the environment can support. Environmental resources are limited. Not all individuals can survive to reproduce.



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A mechanism for evolution: natural selection

Natural selection principle 3: competition and survival of the fittest

Environmental selection pressures such as food availability, predators and some diseases favour those with more advantageous traits/alleles. This may lead to competition between individuals in a population, and those with the advantageous trait may outcompete those without the advantageous trait.

This is known as ‘survival of the fittest’. Those individuals who are more ‘fit’ are better suited to the environment in which the population lives.



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A mechanism for evolution: natural selection

Natural selection principle 4: higher reproductive rate

Individuals with the inheritable advantageous trait are more likely to survive, reproduce and have a higher reproductive rate than those who do not possess the allele.

Natural selection principle 5: heritability

Advantageous alleles are passed to offspring (e.g. camouflaged colouration).



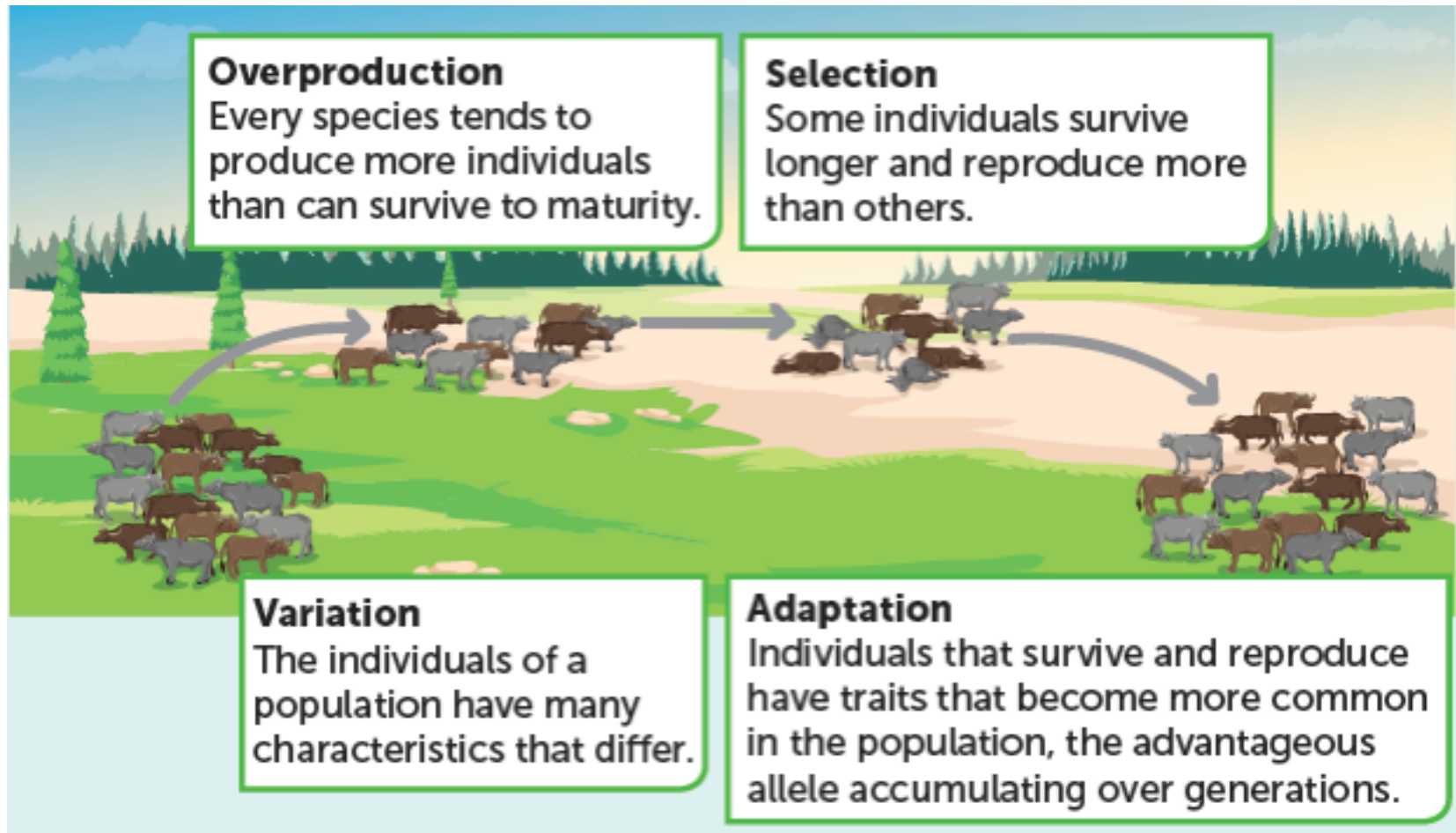
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Natural selection principle 6: allele frequency change

Over consecutive generations, the frequency of the advantageous alleles/traits increases. The frequency of the disadvantageous traits decreases. Over many generations, an advantageous allele can become fixed; the frequency can become 100%. In contrast, the disadvantageous allele can become extinct; the frequency can become 0%.

A mechanism for evolution: natural selection

Natural selection principles: summary



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A mechanism for evolution: natural selection

Natural selection application: giraffes

Here is a case study for a hypothesis about long necks in giraffes being the result of natural selection.

Principle	Application to giraffes
1 Variation	There were alleles for short- and long-necked giraffes due to mutation and sexual reproduction processes.
2 Overproduction	More giraffes are produced than the environmental resources can sustain. There are not enough leaves to feed a whole population, and the last leaves to be eaten are up high.
3 Competition and survival of the fittest	The individuals in the giraffe population compete for the leaves high in the tree. Only those with the advantageous tall allele/trait can reach and consume enough food.

A mechanism for evolution: natural selection

Natural selection application: giraffes

Principle	Application to giraffes
4 Higher productive rate	The individuals who possess the advantageous allele can survive and reproduce and have a higher reproductive rate than those who do not possess the advantageous allele.
5 Heritability	The giraffes with the higher reproductive rate are more likely to pass their inheritable, advantageous allele on to offspring.
6 Allele frequencies change over generations	The next generation of giraffes have a higher frequency of advantageous (tall) alleles and a lower frequency of disadvantageous (short) alleles. Over many generations, the tall allele accumulates until the allele becomes fixed and the disadvantageous short allele becomes extinct.

A mechanism for evolution: artificial selection

Artificial selection: animal and plant breeding

Artificial selection (selective breeding) is the intentional breeding/reproduction of individuals with desirable traits, resulting in changes in allele frequencies in gene pools over time. The traits are beneficial to humans. The breeding for particular traits results in changes in allele frequencies over generations, and therefore this is a mechanism for evolution. Specific allele frequencies will decrease and variation will decrease as humans breed for specific desirable traits.

A mechanism for evolution: artificial selection

Artificial selection: sheep and cattle

Selective breeding in sheep has been practised widely. Selected characteristics include quality and quantity of meat, quality of wool and size of the sheep.

Cows have been bred for meat quality or the quantity or quality of milk they produced. Jersey cows have been bred for both quantity and quality of milk. The Belgian Blue is a breed of cattle that has been bred for the meat industry through artificial selection. The Belgian Blue's unusual physique comes from a naturally occurring 'double muscling' mutation. Due to the mutation, muscle development is not regulated, resulting in huge muscles.



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Artificial selection: fruits and vegetables

Artificial breeding has been applied to many fruits and vegetables, such as corn, bananas and watermelon.

Traits that breeders have tried to incorporate into crop plants include:

- improved quality, such as increased nutrition, improved flavour, seedlessness or greater beauty
- increased yield
- increased tolerance of environmental pressures (salinity, extreme temperature, drought)
- resistance to viruses, fungi and bacteria
- increased tolerance to insect pests
- increased tolerance of herbicides
- longer storage period.

A mechanism for evolution: artificial selection

Artificial selection: fruits and vegetables



Wild banana

The first bananas may have been cultivated at least 7000 years ago in what is now Papua New Guinea, and were stocky and hard, with large, tough seeds throughout the fruit's interior.



Modern banana

Today's tastier bananas are hybrids of two wild banana varieties, *Musa acuminata* and *Musa balbisiana*.



Wild corn

One of the most standout examples of selective breeding is North American sweet corn, which was bred from the barely edible teosinte plant. Natural corn was first domesticated around 7000 BC and was thought to have been as dry as a raw potato.



Modern corn

The corn we eat now is 1000 times bigger and much easier to grow and peel. The majority of the development took place after the 15th century, when European settlers started farming it.



Wild carrot

The first carrots were likely cultivated around the 10th century in Asia Minor and were either white or purple, with thin, forked roots and a strong flavour.



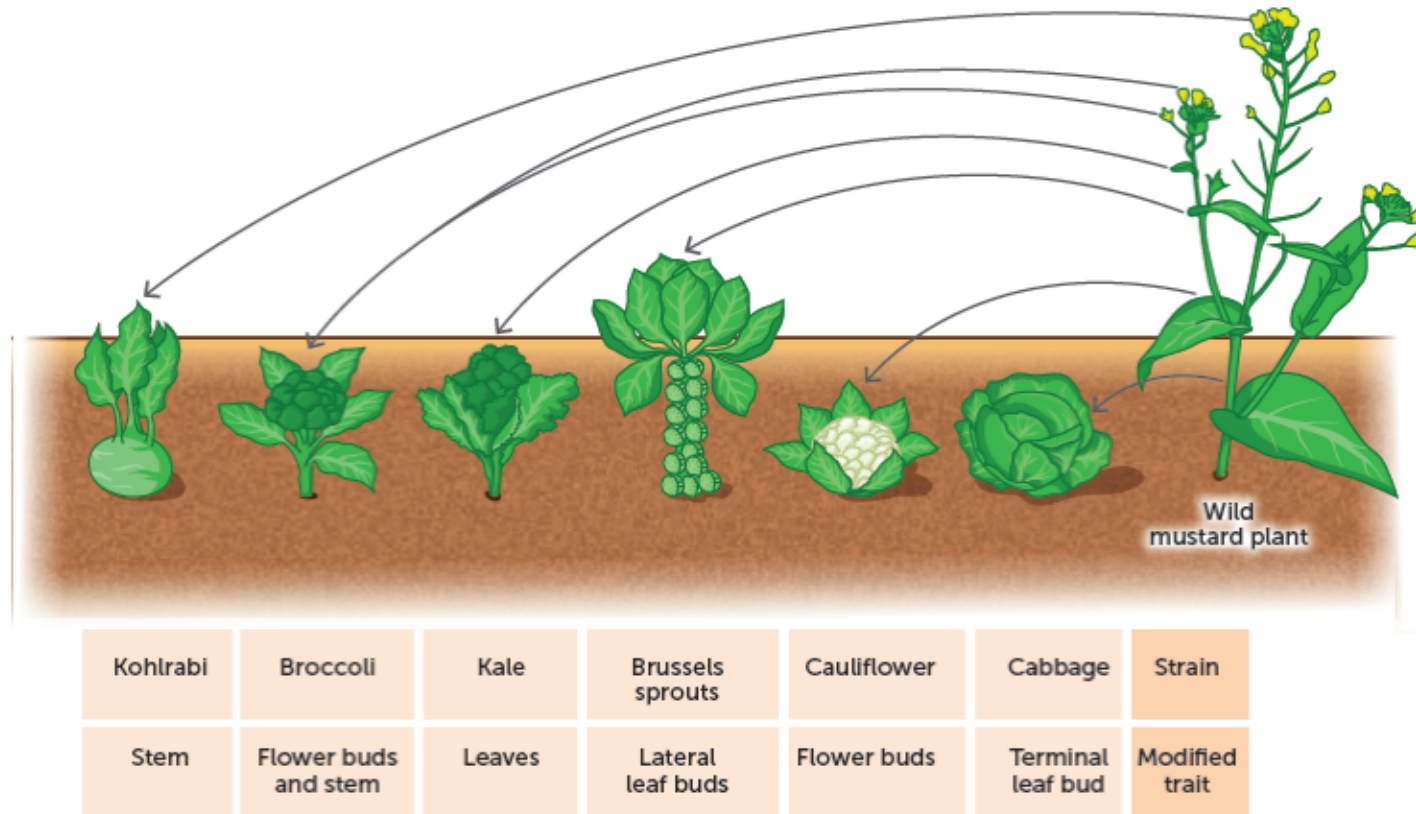
Modern carrot

Carrots today are large, bright orange, and tasty.

A mechanism for evolution: artificial selection

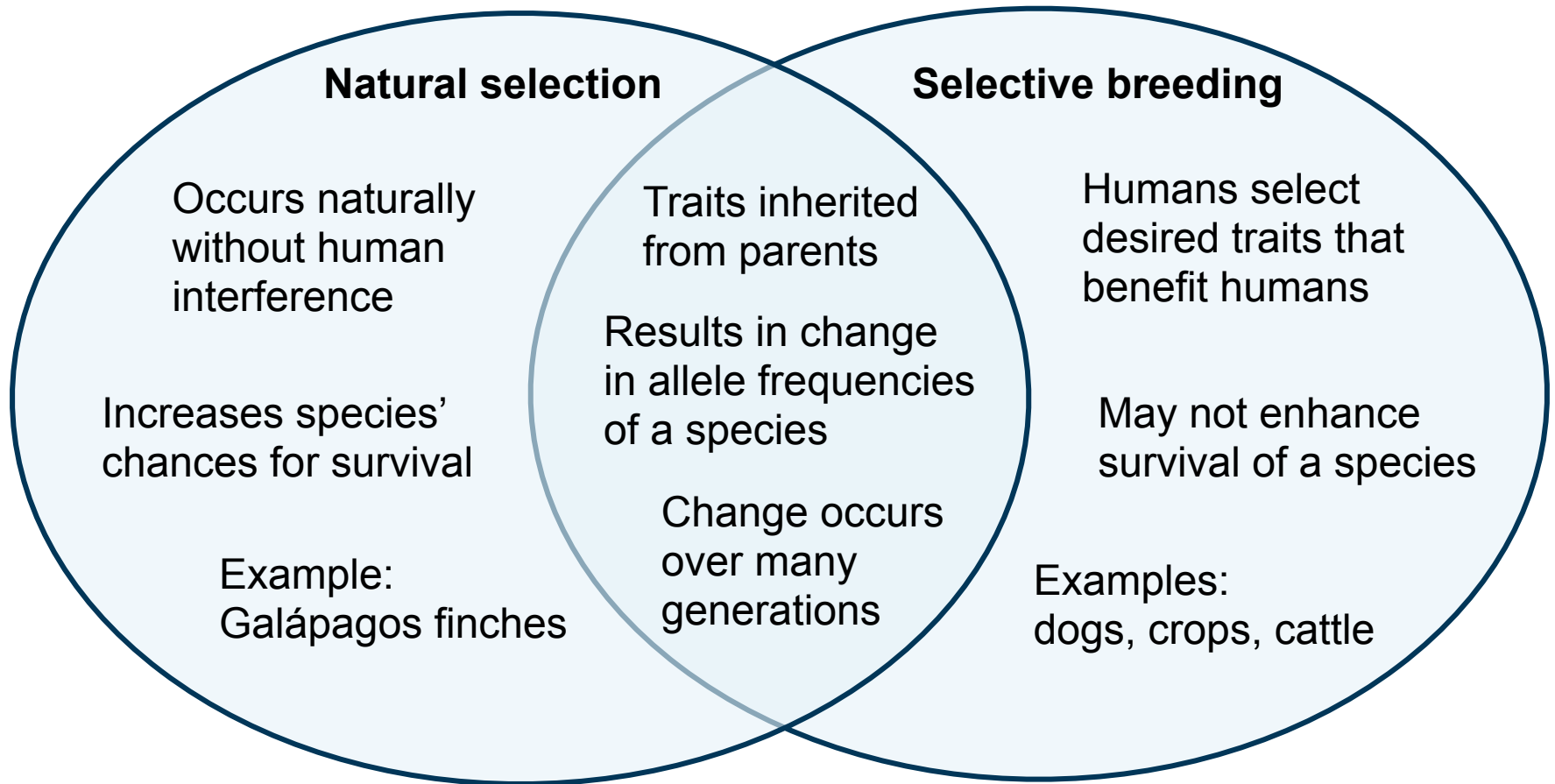
Artificial selection: fruits and vegetables

The wild mustard plant has evolved through selective breeding.



Natural vs artificial selection

Natural selection and selective breeding compared



Natural selection and selective breeding: advantages

Selective breeding	Natural selection
Usually a faster growth rate	Slower growth rate and therefore has time to adapt to changes in the environment, such as poor soil quality
Increased nutritional value, larger yield, pest resistance, drought resistance, disease resistance	Higher genetic variation – less susceptible to changes in the environment. (Artificial selection usually breeds for one trait, which reduces variation; i.e. there is less chance of suitable alleles existing and more chance of extinction.)

A mechanism for evolution: sexual selection

Sexual selection

Sexual selection is a selection process by male and female individuals of a population for an inherited trait that assists in copulation or in the winning of a mate. It describes a form of selection in which individuals with certain inherited characteristics or behaviours are more likely than others to obtain mates/ copulate and pass on their genes.

Sexual selection can produce quite spectacular effects, such as the enormous antlers of a moose, or the long, showy tail of a male peacock.

Over many generations, the frequency of the advantageous allele increases. Therefore, this process is a mechanism for evolution.

Although sexual selection is a type of natural selection, the advantageous trait does not necessarily assist the individual to survive its environmental selection pressure; it only helps win a mate.

A mechanism for evolution: sexual selection

Sexual selection example:

The lyrebird is one of the most accomplished mimics in the animal kingdom. The males sing complex songs, mimicking animal and bird sounds, and even mechanical sounds, such as chainsaws. The males with a greater repertoire achieve better reproductive success.



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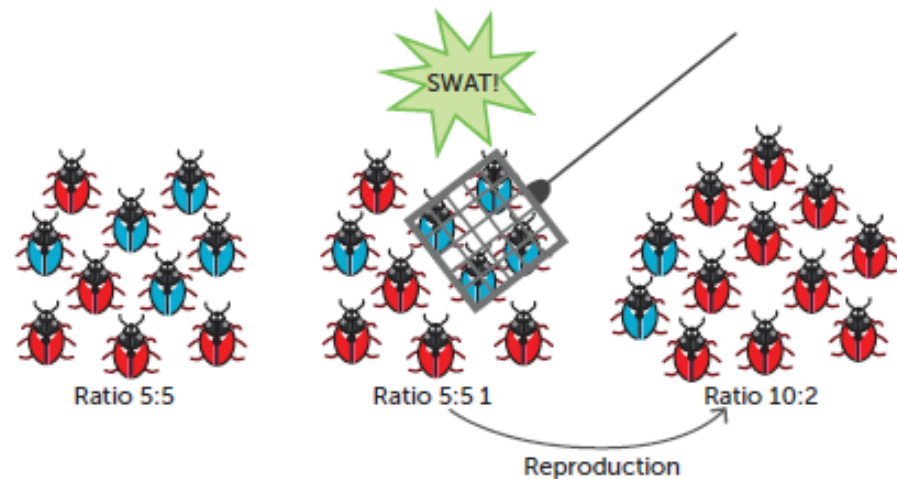
A mechanism for evolution: genetic drift

Genetic drift

Random changes from generation to generation are known as **genetic drift**. Genetic drift is a mechanism of evolution in which allele frequencies of a population change over generations due to chance.

There is usually a loss of genetic variation over generations.

Genetic drift occurs in all populations, but its effects are strongest in small populations. The smaller the population, the faster the fixation of alleles and the extinction of other alleles can occur.



A mechanism for evolution: genetic drift

Genetic drift: bottleneck effect

The bottleneck effect occurs when there is a disaster of some sort that reduces a population to a small handful, which rarely represents the actual genetic makeup of the initial population. This leaves smaller variation among the surviving individuals.

Example:

Cheetahs are an endangered species that have survived a drastic genetic bottleneck. Facing a declining population, the surviving parents mated with their own offspring, and the resulting generations were left with strikingly similar alleles. One of these is a mutated allele with negative effects on fertility. Other shared alleles result in lowered resistance to disease.



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A mechanism for evolution: genetic drift

Genetic drift: founder effect

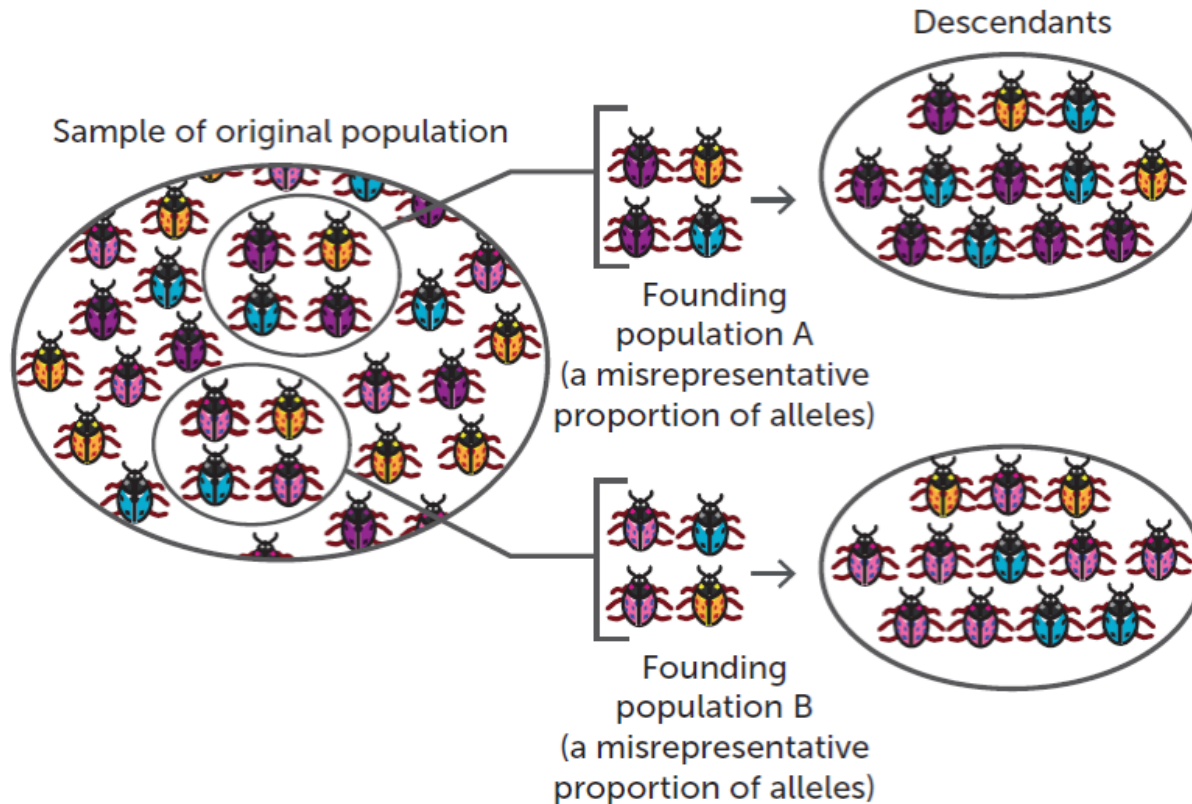
A few individuals who move to a new area and become isolated from a larger population might not carry all the alleles that were present in the original population. This means that the isolated population has less genetic diversity than the original population, and deleterious recessive alleles may have a higher chance of coming together than they did in the original population.

The **founder effect** happens when there is a dramatic decrease in genetic diversity caused by the development of small colonies of individuals, sourced from the original population, that remain isolated from other colonies.

Only a small subset of the genetic diversity of the source population is likely to be included in the new population, and the relative frequencies of these alleles may be very different from what they were in the source population.

A mechanism for evolution: genetic drift

Genetic drift: founder effect



A mechanism for evolution: gene flow

Gene flow

Gene flow is transfer of alleles, and it results from the migration of individuals from one population to another. This can be due to immigration (individuals joining a population), and emigration (individuals leaving a population).



Micro-evolution and macro-evolution explained

A change in the frequencies of various alleles within a population is a process called **micro-evolution**, which is change below the species level.

Macro-evolution is a set of processes that attempt to explain how new species have evolved in such large numbers. There are three broad processes that work together towards macro-evolution.

1 Natural selection favours phenotypes that make the population better adapted to its environment. Populations change over time as their gene pools accumulate small changes in response to natural selection. This is called micro-evolution.

2 Eventually a population accumulates so many changes that a new species can be identified. This process can lead to speciation, the multiplication of species.

3 Sometimes a rapid series of speciation events leads to the development of a whole collection of new species, or even genera, families, or higher classification groups. This is referred to as macro-evolution.

Micro-evolution and macro-evolution

Comparison of micro-evolution and macro-evolution

Microevolution	Macroevolution
A change in the frequencies of various alleles within a population	Changes in allele frequencies in more than one population/species
Change below the species level	Major evolutionary changes above the species level
Small-scale change in the gene pool of a population due to the mechanisms of mutation, natural selection, genetic drift and gene flow	Large scale change resulting from an accumulation of micro-evolutionary changes over many generations and a very long time

Species terminology

A **species** is a group of organisms that can interbreed to produce viable, fertile offspring and cannot breed with the individuals of another species to produce fertile offspring.

This is the **biological species concept** – a genetically isolated group with its own gene pool.

The **morphological species concept** defines a species by structural features. Individuals of the same species are morphologically similar.

Speciation is the formation of a new species. It is the process of one species splitting into two or more species.

Speciation

Speciation example:

The famous Galápagos tortoises are similar to the much smaller Chaco tortoise (*Geochelone chilensis*), found in South America, but are completely separate species. Darwin hypothesised that the tortoises on the islands originally came from the mainland population, but had changed over time to become better suited to the environment of the Galápagos.



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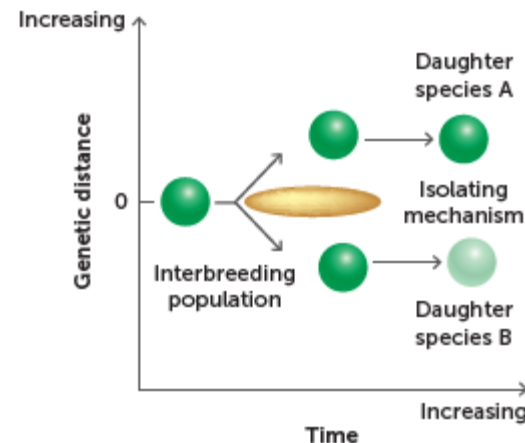
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Mechanisms of speciation: isolating mechanisms

Isolating mechanisms

Isolating mechanisms separate two subgroups of a population and prevent them from producing fertile, viable offspring. These mechanisms can operate before reproduction has occurred or after reproduction.

The organisms become so genetically diverse that they form two new species. They are then no longer able to interbreed, even if the populations come back together.



Mechanisms of speciation: isolating mechanisms

Pre-reproductive isolating mechanisms

Pre-reproductive isolating mechanisms are biological or ecological mechanisms that prevent organisms from being able to interact to reproduce:

- Temporal (time) mechanisms: individuals breed during different seasons of the year or times of the day
- Behavioural mechanisms: individuals have different courtship patterns
- Morphological mechanisms: individuals have different reproductive structures



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Example:

Some periodical cicadas (genus *Magicicada*) hatch every 17 years; others hatch every 13 years. The unusually lengthy life cycles may act to prevent different populations interbreeding and producing hybrid offspring.

Post-reproductive isolating mechanisms

Post-reproductive isolating mechanisms do not prevent mating from occurring but they do prevent young from being produced:

- Gamete mortality: the gametes do not survive
- Zygote mortality: the zygote forms but does not survive
- Hybrid sterility: adult offspring are formed but are infertile because they are unable to produce viable gametes, usually because of having received a different number or type of chromosomes from each species.

Allopatric speciation

In allopatric speciation (from the ancient Greek 'allos' = other and 'patra' = homeland), gene flow is disrupted when populations become physically separated through geographical isolation. The populations diverge. This may be because of different selection pressures acting on the two populations, or it may be due to other random processes such as genetic drift.

Physical barriers that can separate a subpopulation from its original population:

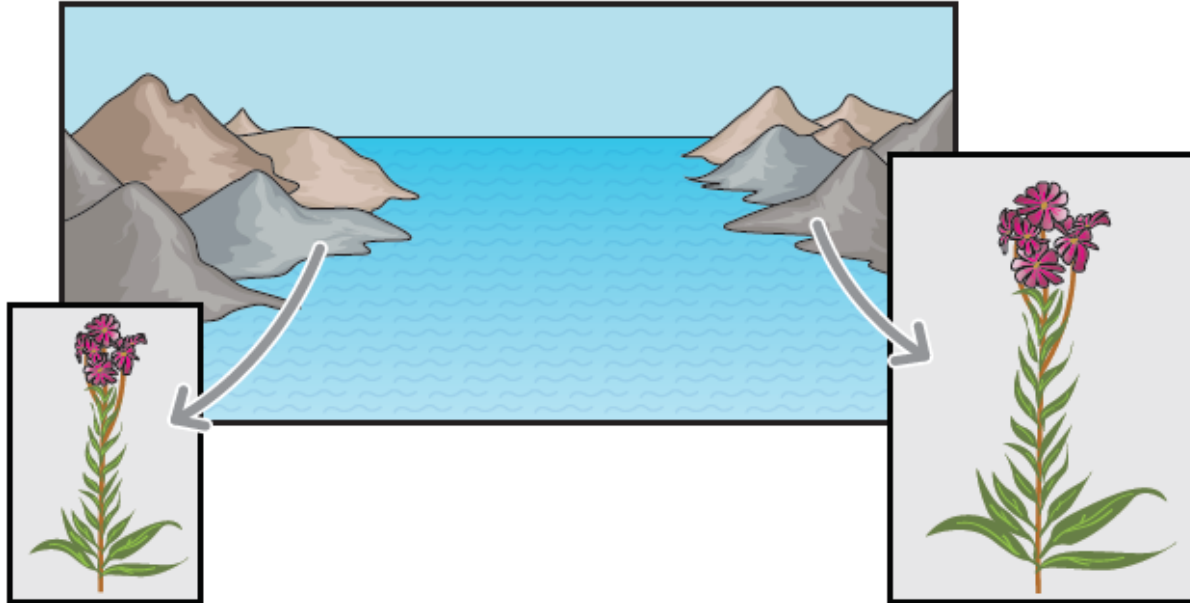
- water, for terrestrial organisms
- land, for aquatic organisms
- mountains.

New physical barriers can arise due, among other things, to:

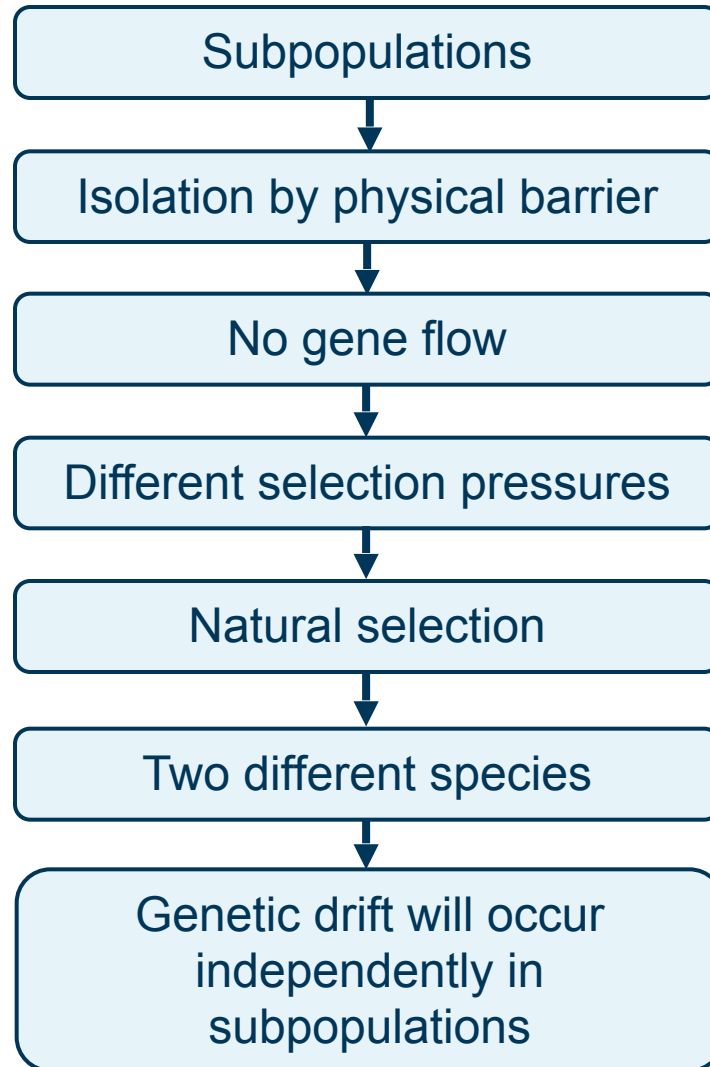
- continental drift
- rising sea levels
- climate change.

Allopatric speciation

A physical barrier such as the rising of sea level separates two populations and prevents gene flow. Populations adapt to differing environments on opposite sides of the barrier (natural selection). Genetic drift causes different random changes in both subpopulations.



Steps in allopatric speciation



Australian bushfires and species extinction

More than 1 billion animals were killed in the 2019/2020 Australian bushfire season. Endangered species, including the long-footed potoroo, Kangaroo Island's glossy black-cockatoo and Batemans Bay's spring midge orchid were pushed towards extinction.



Getty Images/Lisa Marce Williams

The devastating wildfires undid decades of careful conservation work on Kangaroo Island and have threatened to wipe out some of the island's unique fauna altogether.

Preventing extinction by preserving genetic diversity

Populations with reduced diversity face increased risk of extinction, so conservation projects usually focus on maintaining genetic diversity.

Rapid extinction events can lead to greater loss of large organisms than of small ones. A large distribution area is generally a big advantage, because it may allow some pockets of habitat to survive.

Large population size can also be some protection, because the population is likely to have a more diverse gene pool and thus a greater variety of alleles and phenotype options as the pressures from natural selection change.